

Citrus Leaf Disease Classification

Deep Learning Model Comparison and Evaluation Report

Prepared from experiments on the supplied citrus leaf dataset.

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1. Executive Summary

This project evaluates seven pretrained image-classification architectures for five citrus leaf classes: Black spot, Canker, Greening, Healthy, and Melanose. Among the recorded runs, ViT-B/16 achieved the highest test accuracy at 88.76%, with macro F1 0.8554 and weighted F1 0.8905.

2. Dataset Description

The recorded dataset contains 1,716 images. The class distribution is summarized below.

Class	Images	Percentage
Black spot	238	13.87%
Canker	237	13.81%
Greening	220	12.82%
Healthy	705	41.08%
Melanose	316	18.41%

The dataset is imbalanced: Healthy has the largest number of images. Per-class precision, recall, and F1 should be considered alongside overall accuracy.

3. Methodology

The recorded experiments use pretrained architectures with modified classification heads. The repository-based pipeline used image resizing/cropping, online training augmentation, ImageNet normalization, weighted cross-entropy with label smoothing, and an adaptive learning rate scheduler. The summarized runs used eight epochs.

ViT-B/16, MobileNetV3-Large, ConvNeXt-Tiny, Swin-Tiny, and MobileViT-Small used the same saved stratified split with seed 42. ResNet34 and EfficientNet-B0 came from earlier experiments with different splits.

4. Model Comparison

The table reports best validation accuracy, test accuracy, macro F1, weighted F1, and recorded training time. Missing times are left as N/A.

Model	Val. Acc.	Test Acc.	Macro F1	Weighted F1	Min.
ResNet34	70.43%	76.74%	0.7500	0.7800	N/A
EfficientNet-B0	86.38%	88.37%	0.8500	0.8800	N/A
ViT-B/16	88.33%	88.76%	0.8554	0.8905	7.13
MobileNetV3-Large	73.15%	77.52%	0.7459	0.7849	1.16
ConvNeXt-Tiny	81.71%	85.27%	0.8269	0.8586	1.39
Swin-Tiny	85.21%	85.27%	0.8222	0.8566	1.33
MobileViT-Small	78.60%	81.40%	0.7731	0.8137	1.24

Comparison caveat: ResNet34 and EfficientNet-B0 used different splits from the five shared-split models. Their results are preliminary and should not be treated as a fully controlled ranking.

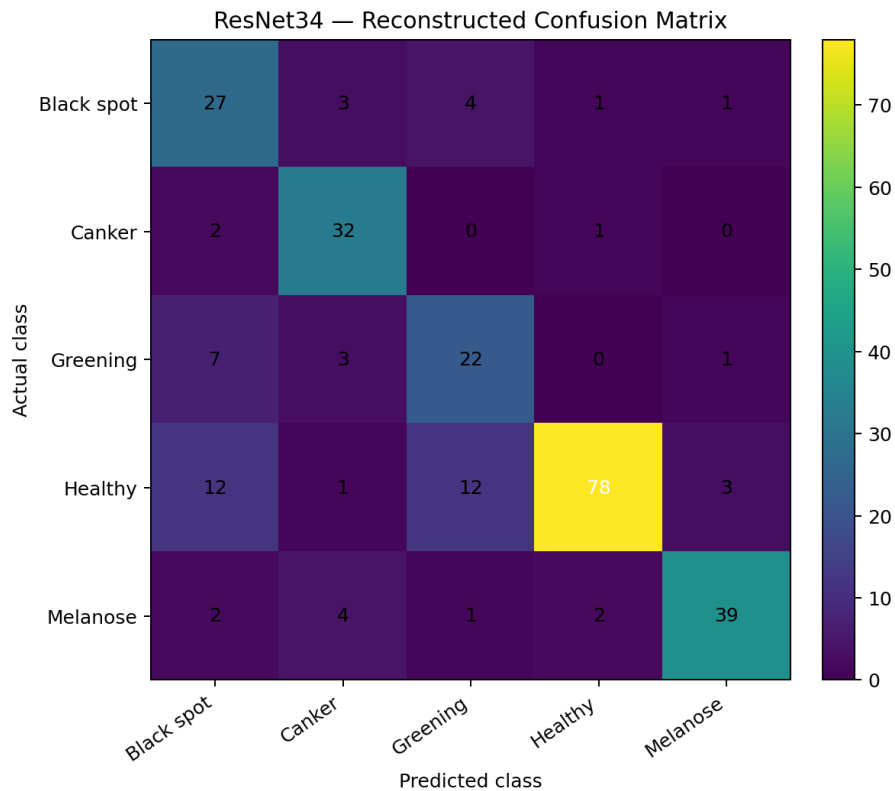
5. Interpretation of Results

ViT-B/16 has the highest recorded test accuracy in this collection of runs. EfficientNet-B0 also recorded strong accuracy, but the split difference limits direct comparison.

MobileViT-Small achieved 81.40% test accuracy, with Black spot recall of 0.4444. Many Black spot test images were therefore misclassified. This supports evaluating disease classes individually rather than relying only on aggregate accuracy.

6. Confusion Matrices and Classification Reports

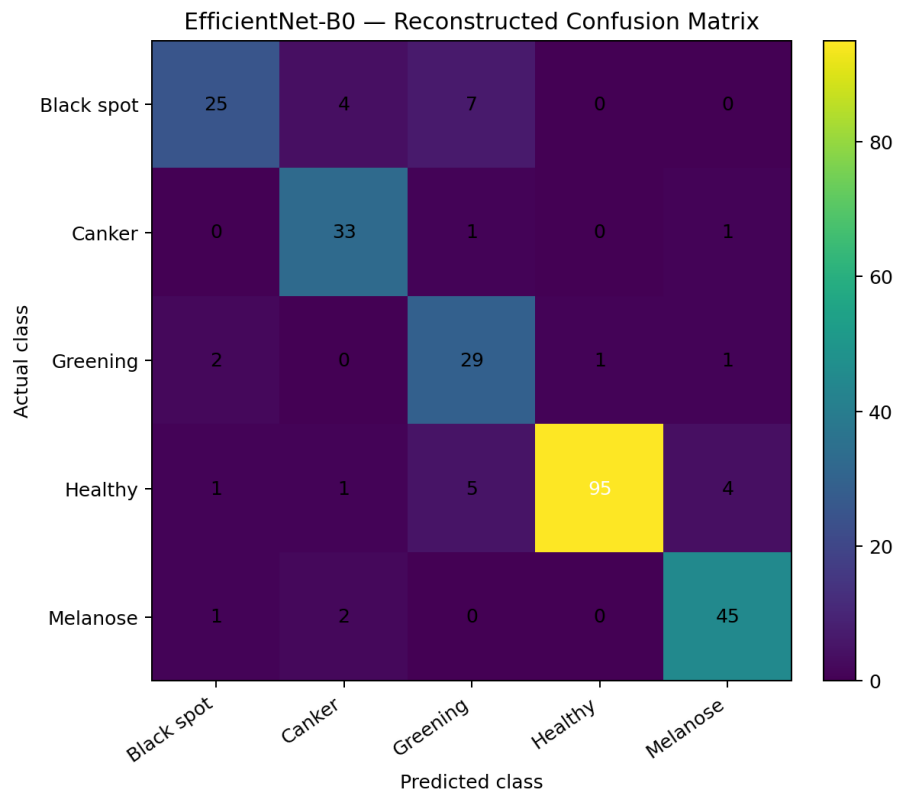
ResNet34



Classification report:

	precision	recall	f1-score	support
Black spot	0.54	0.75	0.63	36
Canker	0.74	0.91	0.82	35
Greening	0.56	0.67	0.61	33
Healthy	0.95	0.74	0.83	106
Melanose	0.89	0.81	0.85	48
accuracy			0.77	258
macro avg	0.74	0.78	0.75	258
weighted avg	0.80	0.77	0.78	258

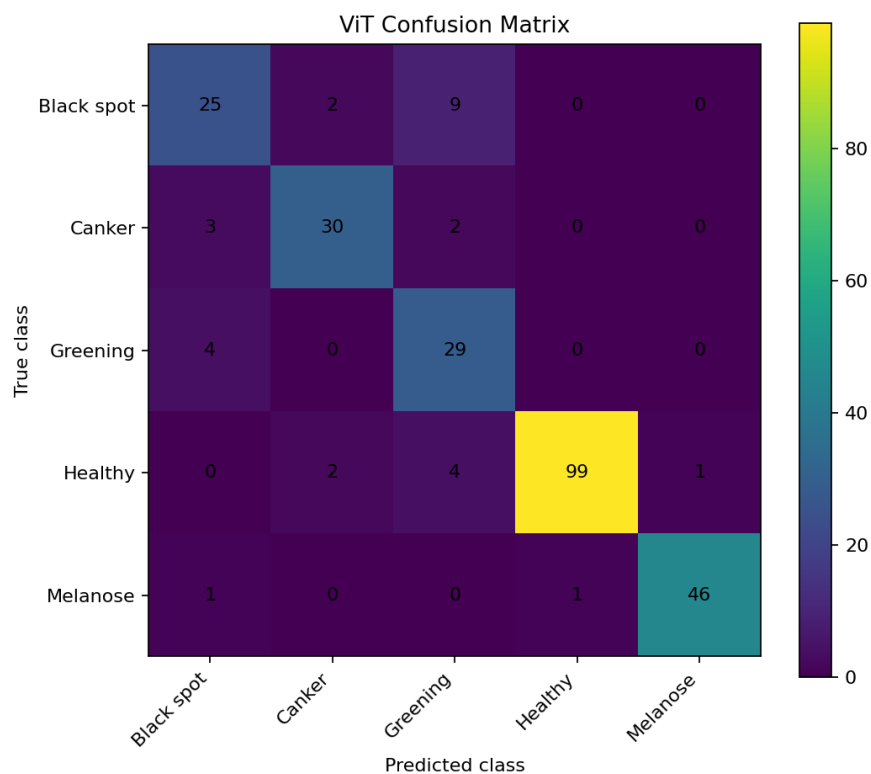
EfficientNet-B0



Classification report:

	precision	recall	f1-score	support
Black spot	0.86	0.69	0.77	36
Canker	0.82	0.94	0.88	35
Greening	0.69	0.88	0.77	33
Healthy	0.99	0.90	0.94	106
Melanose	0.88	0.94	0.91	48
accuracy			0.88	258
macro avg	0.85	0.87	0.85	258
weighted avg	0.89	0.88	0.88	258

ViT-B/16



Classification report:

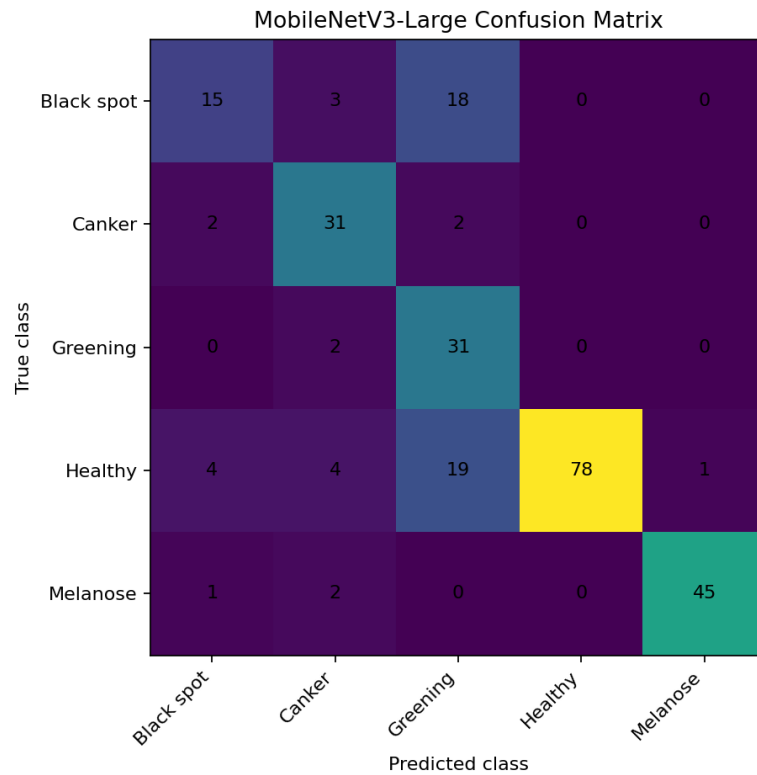
Best validation accuracy: 0.883268
 Test accuracy: 0.887597
 Training duration seconds: 427.85

	precision	recall	f1-score	support
Black spot	0.7576	0.6944	0.7246	36
Canker	0.8824	0.8571	0.8696	35
Greening	0.6591	0.8788	0.7532	33
Healthy	0.9900	0.9340	0.9612	106
Melanose	0.9787	0.9583	0.9684	48

accuracy			0.8876	258
macro avg	0.8535	0.8645	0.8554	258
weighted avg	0.8985	0.8876	0.8905	258

Confusion matrix:
 [[25 2 9 0 0]
 [3 30 2 0 0]
 [4 0 29 0 0]
 [0 2 4 99 1]
 [1 0 0 1 46]]

MobileNetV3-Large



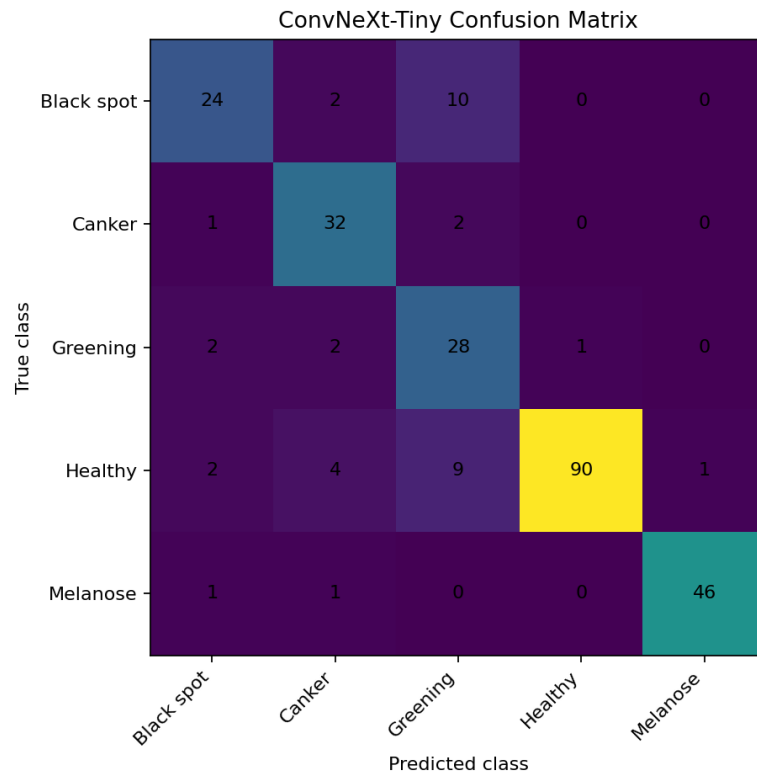
Classification report:

Best validation accuracy: 0.731518
 Test accuracy: 0.775194
 Training duration seconds: 69.49

	precision	recall	f1-score	support
Black spot	0.6818	0.4167	0.5172	36
Canker	0.7381	0.8857	0.8052	35
Greening	0.4429	0.9394	0.6019	33
Healthy	1.0000	0.7358	0.8478	106
Melanose	0.9783	0.9375	0.9574	48
accuracy			0.7752	258
macro avg	0.7682	0.7830	0.7459	258
weighted avg	0.8448	0.7752	0.7849	258

Confusion matrix:
 [[15 3 18 0 0]
 [2 31 2 0 0]
 [0 2 31 0 0]
 [4 4 19 78 1]
 [1 2 0 0 45]]

ConvNeXt-Tiny



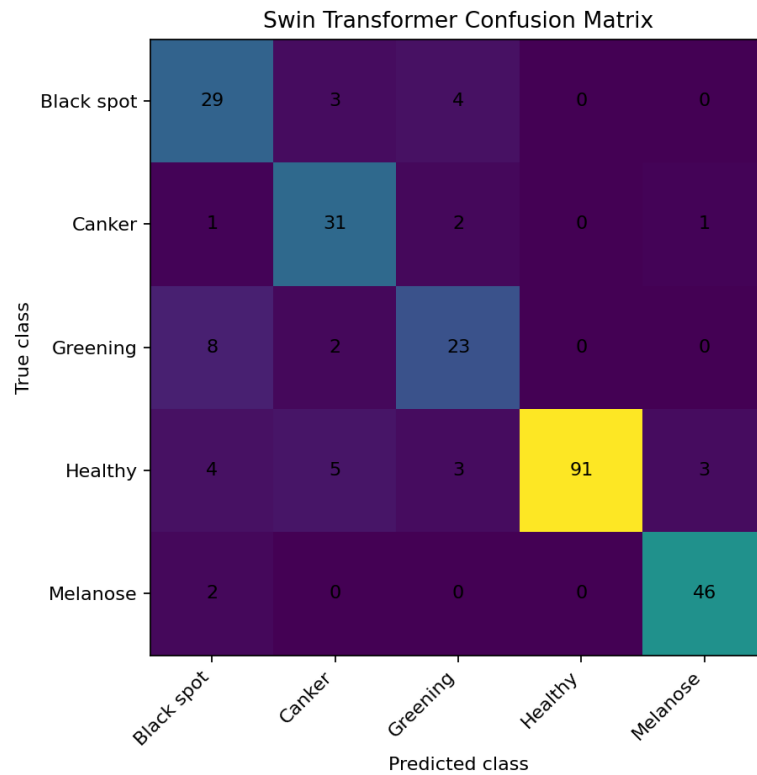
Classification report:

Best validation accuracy: 0.817121
 Test accuracy: 0.852713
 Training duration seconds: 83.53

	precision	recall	f1-score	support
Black spot	0.8000	0.6667	0.7273	36
Canker	0.7805	0.9143	0.8421	35
Greening	0.5714	0.8485	0.6829	33
Healthy	0.9890	0.8491	0.9137	106
Melanose	0.9787	0.9583	0.9684	48
accuracy			0.8527	258
macro avg	0.8239	0.8474	0.8269	258
weighted avg	0.8790	0.8527	0.8586	258

Confusion matrix:
 [[24 2 10 0 0]
 [1 32 2 0 0]
 [2 2 28 1 0]
 [2 4 9 90 1]
 [1 1 0 0 46]]

Swin-Tiny



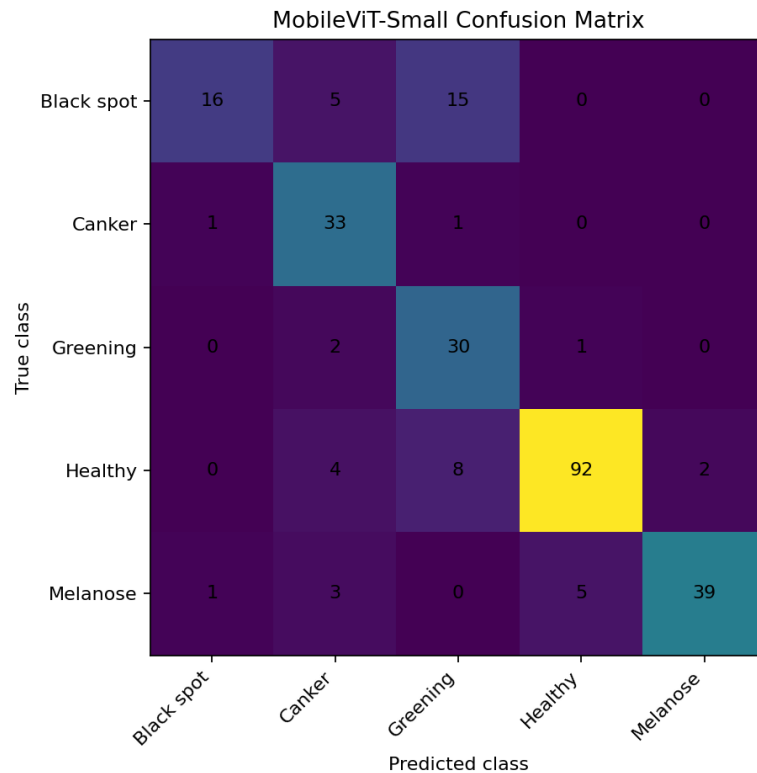
Classification report:

Best validation accuracy: 0.852140
 Test accuracy: 0.852713
 Training duration seconds: 79.93

	precision	recall	f1-score	support
Black spot	0.6591	0.8056	0.7250	36
Canker	0.7561	0.8857	0.8158	35
Greening	0.7188	0.6970	0.7077	33
Healthy	1.0000	0.8585	0.9239	106
Melanose	0.9200	0.9583	0.9388	48
accuracy			0.8527	258
macro avg	0.8108	0.8410	0.8222	258
weighted avg	0.8685	0.8527	0.8566	258

Confusion matrix:
 [[29 3 4 0 0]
 [1 31 2 0 1]
 [8 2 23 0 0]
 [4 5 3 91 3]
 [2 0 0 0 46]]

MobileViT-Small



Classification report:

Best validation accuracy: 0.785992
 Test accuracy: 0.813953
 Training duration seconds: 74.59

	precision	recall	f1-score	support
Black spot	0.8889	0.4444	0.5926	36
Canker	0.7021	0.9429	0.8049	35
Greening	0.5556	0.9091	0.6897	33
Healthy	0.9388	0.8679	0.9020	106
Melanose	0.9512	0.8125	0.8764	48
accuracy			0.8140	258
macro avg	0.8073	0.7954	0.7731	258
weighted avg	0.8530	0.8140	0.8137	258

Confusion matrix:
 [[16 5 15 0 0]
 [1 33 1 0 0]
 [0 2 30 1 0]
 [0 4 8 92 2]
 [1 3 0 5 39]]

7. Limitations

- ResNet34 and EfficientNet-B0 used different splits from the five shared-split experiments.
- Eight training epochs do not establish the maximum attainable performance.
- The dataset is imbalanced, so accuracy alone may hide class-specific weaknesses.
- Performance on external field images has not been established by these results.
- The original ResNet34 and EfficientNet-B0 folders lacked saved training histories. Their confusion matrices were reconstructed from previously recorded count matrices.

8. Recommendations

- Retrain all seven architectures with exactly the same data split and evaluation procedure for a fair comparison.
- Compare macro F1, class-wise recall, confusion matrices, and model size in addition to accuracy.
- Investigate Black spot and Greening errors and inspect ambiguous samples.

- Evaluate the selected model on independent field images before real-world use.

9. Conclusion

The recorded experiments show promising performance for citrus leaf classification on this dataset. ViT-B/16 produced the highest recorded test accuracy among these runs. A controlled comparison using the same split for every model and evaluation on independent field images is recommended before choosing a deployment model.

10. Package Contents

The package includes model checkpoints, available classification reports, confusion matrices, test predictions, training histories, class mappings, a consolidated comparison table, dataset class distribution, and this report in PDF and editable DOCX formats.